

HYDROLOGY

Self
confidence

Achieve
Daily Target

Multiple
revision

To The Point-ByDhyanPal(ESE'17AIR-179,GATE'18AIR-93,GATE'16AIR-145)



HYDROLOGY EXPRESS

- ① catchment area → planimeter (on topographic map)
• overlay grid (small squares sheet)
- ② capillary potential → Tensiometer
- ③ Transpiration → phytometer
- ④ evaporation → by dalton
→ lysimeter, field plot
- ⑤ evapotranspiration → A transducer
- ⑥ Infiltration → flooding type infiltrometer, rainsimulator, hydrograph analysis.
- ⑦ continuous record of temp. & humidity → Thermohygrograph
- ⑧ speed of wind → Anemometer
- ⑨ Humidity → Hygrometer
- ⑩ Relative Humidity → psychrometer
- ⑪ solar radiation → pyrheliometer
- ⑫ Rainfall → rain gauge, pluviometer, ombrometer, dry weather meter
- ⑬ Isoleth → line joining same PET (or evapotranspiration)
- ⑭ PET → Thornthwaite formula, Penman eqn
(mass transfer, energy balance)
- ⑮ AET → By Thornthwaite water balance eqn, Lacey field plot
- ⑯ mass curve → accumulated rain vs time $\frac{\Delta P}{\Delta t}$ → To the point
- ⑰ missing rainfall → normal ratio method ($> 10\%$)
- ⑱ double mass curve → inconsistency check
- ⑲ Isohyet → line joining points of equal rainfall magnitude
- ⑳ Isopleth → equal depth of rainfall
- ㉑ area velocity method → stage discharge curve constant
- ㉒ slope area method → energy conservation based
(in this sectional elevation known)
- ㉓ moving boat method → for wide stream having
(special case of Area-velocity method) unsteady flow condition
- ㉔ Tipping bucket method → intensity of rainfall
- ㉕ steven weighing bucket & mass curve
natural syphon *
- ㉖ flow mass = plot of cumulative runoff volume vs time
- ㉗ flood duration curve / discharge frequency curve
→ Q vs time flood exceeded
- ㉘ hydrograph = stream flow vs time in chronological order

- ㉙ S curve → Rain intensity 1 cm/hr slope ordinate of IUH
- ㉚ UH → Sherman (1932) → based on system approach
- ㉛ ϕ → max. infiltration rate
- ㉜ linear reservoir = storage is fn of out flow discharge
- ㉝ Linear channel = no Attenuation, only Lag
- ㉞ Hydrologic routing → $S = f^n(I, Q)$ → based on only continuity eqn
↳ Muskingum eqn
- ㉟ Hydraulic routing → continuity eqn + eqn of motion of unsteady flow
- ㊱ Synthes → Synthetic UH → ungauged basin
↳ Lag time → peak to mid rain
But in UH → C to mid rain
- ㊲ envelop discharge curve → log-log curve
discharge vs catchment area
- ㊳ Coriolis force → river meandering → { cause by Earth rotation }
- ㊴ effect → velocity distribution
- ㊵ sensible heat → bowen ratio $\left(\frac{\text{heat diffusivity}}{\text{vapour}} \right)$
- ㊶ soil conservation measure → reduction in peak in small & medium flood
- ㊷ Urbanisation → Increased peak in flood hydrograph
- ㊸ Khosla curve → design of weir & barrage on permeable foundation
- ㊹ Coriolis coeff. / KE correction factor → used to account for effect of non uniform velocity distribution
- ㊺ Storativity / storage coefficient → volume of water given unit plan area of Aquifer when piezometric surface fall by unity.

Term	line joining equal
Isobar	pressure
Isobath	Depth in sea
Isobront	thunderstorm at same time
Isohaline	salinity
Isoneks	Sunshine
Isorif	snowfall amount
Isosyme	frost
Isotherm	Temp.

To the point
dhyans
21/4/2020

हाइड्रोलॉजी रामबाण - By ध्यान पाल

Ocean → evaporation > precipitation
 land → " < "

cyclone	anticyclone

frontal = warm + cold air mass
 cyclone = low pressure
 anticyclone = high pressure
 convective = Temp. difference
 orographic = barrier

Hydro-meteorological station :- (1) Rain gauge (rec. ordinary)
 (2) Thermometer (3) anemometer
 (4) wind direction indicator (5) sunshine recorder
 (6) Pan evaporation (7) thermohygraph.

- evaporation → day + night { continuous process }
- Transpiration → during only day time (during photosynthesis)
 transpiration rate depend on growth of plant
- Interception → short term retention of rain by leaves of vegetation.

change in storage (ΔS) = Inflow - outflow
 $\Delta S = P - (R + E + T + \text{net groundwater flow})$

Residence time ⇒ Avg. duration of particle of water to pass through a phase of hydrological cycle.
 Order :- Ocean > Global groundwater > River > Atm. vapour
note :- if residence time ↑ ⇒ difficult to predict that phase.

Precipitation forms :- All forms of water-reaches to earth from atmosphere.
 major → rain, snow

① Rain	(0.5-6 mm) size (>6mm break up) • light rain < 2.5 mm/hr • moderate 2.5-7.5 • heavy > 7.5 • in India mainly rain by orographic phenomenon • rain measured by rain gauge / pluviometer / ombrometer
② Snow	0.06-0.15 gram/cc (avg = 0.10 gm/cc) ice crystal

rainy day
 $> 2.5 \text{ cm}$
 day

③ Drizzle	Intensity < $1 \frac{\text{mm}}{\text{hr}}$, size < 0.5 mm
④ Glaze	when rain & drizzle comes in contact with cold ground (0°C), water drop freeze to form an ice coating known as Glaze.
⑤ Sleet	• frozen rain drops of transparent grain. • forms when rainfall through air at subfreezing temp (0°C)
⑥ Hail (Showery ppt.)	• size > 8 mm • occurs in violent thunderstorm in which vertical current are very strong.

ideal distribution $\left\{ \begin{array}{l} \text{no Temporal (wrt time)} \\ \text{no spatial (wrt space)} \end{array} \right.$
 In India ⇒ spatiotemporal variation

Coefficient of variation (Cv) = $\frac{\sqrt{n-1}}{P} \times 100$
 • Avg. value of annual rainfall ⇒ 35-40 yr (20 year Avg.)
 • In India Annual Average rainfall ⇒ 118.3 cm.

Index of wetness = $\frac{\text{Rainfall in a given year}}{\text{Avg. Annual rainfall}} \times 100$
 { can be > 100% }

100% ⇒ normal year (monsoon normal)
 < 100% ⇒ Bad year
 > 100% ⇒ good year

Rainfall deficiency = 100 - Index wetness

Drought :-

1. meteorological Drought	• due to > 25% deficiency in ppt from normal Impact ⇒ water cycle disturb (imbalance)
2. Hydrological drought	if meteorological drought continues for long time. Impact ⇒ (i) GWT ↓ (ii) drying of lake, river, stream (iii) reduction of water supply
3. Agriculture drought ↓ denote by Aridity index	• Deficiency of water availability for-plant consumption Impact :- Reduction in crop yield.

drought year $\Rightarrow$ affected area $> 20\%$ of total area
 drought prone area $\Rightarrow$ drought prob. $\Rightarrow 0.2-0.4$
 chronically drought $\Rightarrow$ serious (70-90%)
 prone area

Agriculture drought $\rightarrow$ denote by Aridity Index (AI)

$$AI = \frac{PET - AET}{PET} \times 100$$

$\rightarrow$ {calculated/published Biweekly Basis}

- AET $\rightarrow$ calculated by Thorn's water balance technique
- PET $\rightarrow$ Blaney Piddle formula, Penman eqn
 $\rightarrow$ equal PET (Isoleth)

other's Index to show Agriculture drought

- 1- moisture availability Index (MAI)
- 2- Palmer Index (Palmer drought Index)

Raingauge Installation condition :-

- 1- Levelled surface (nondulation)
 { to reduce wind effect }
 - 2- fencing for raingauge 5.5m x 5.5m
 - 3- put at sufficient height to prevent splashing, flooding
 - 4- no object should be near to R.G. than
 { 30 meter, 2x height of obstruction }
- So that water enter into R.G. without any obstruction

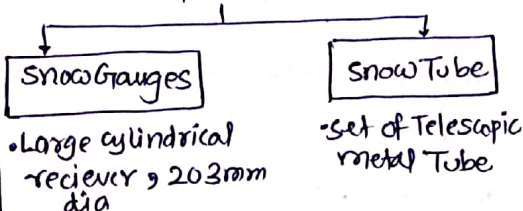


Raingauges :-

1. Non-recording - Symon's Gauge

- 127 mm diameter
- 8:30 AM daily reading taken
- can measure snow fall also

Water equivalent of snow



note:- Antifreezing Agent $\rightarrow$ to melt snow

2- recording Raingauges :- continuous plot of rainfall vs time

① Tipping Bucket

• Hyetograph $\rightarrow$ $\frac{cm}{hr}$
 • pair of bucket when rainfall of 0.25mm, collect in 1 bucket, it tips & brings the other one in position

② Steven Weighing Bucket

• mass curve
 for All kind of ppt. (rain, snow, sleet)

③ Natural syphon or float type

• mass curve
 It is standard recording R.G. in India

• vertical line suddenly empty of float chamber by syphon action, which reset the pen to zero level

latest Rain Gauge Technology :-

① Telemetry raingauge

- recording type R.G.
- for remote & inaccessible location.
- can transfer rainfall data to base station.

② Radar-Based R.G.

- Based for finding areal extent, location & movement of rainstorms.

R.G. density :- WMO :- (min 10% Recording type R.G.)

Plain	1 station per 520 km ²
moderate elevation (upto 400m)	260-390
Hilly area with heavy rainfall	130 km ²

overall India
 1/600 km²
 Israel
 37/1000 km²

19/4/2020

$$N = \left(\frac{C_v}{\epsilon}\right)^2 \quad C_v = \frac{\sigma_{\bar{p}}}{\bar{p}} \times 100 \quad \sigma_{\bar{p}} = \sqrt{\frac{\sum (p_i - \bar{p})^2}{n-1}}$$

Statistical method (uses σ & solve problem)

$$\epsilon_{\text{allowable error}} = \frac{\text{error}}{\text{mean rainfall in } \bar{p}} \times 100$$

Preference of R.G. :- (i) जहाँ पहले से कोई R.G. न हो
(ii) जहाँ पहले से rain gauge density कम हो

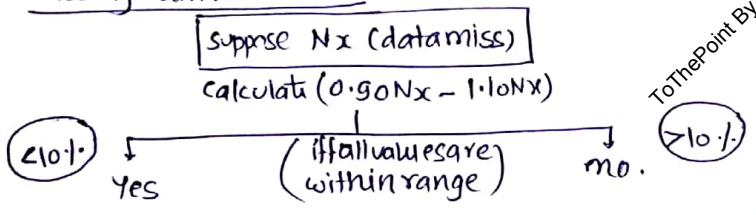
Normal Rainfall :- over a specified period of 30 year [recomputed every decade]

used in missing rainfall data calculation.

Ex. (1981-2010) (1991-2020)
for 2019 we use 1981-2010 data.

Preparation of data $\left\{ \begin{array}{l} \text{continuity check} \\ \text{(to check missing rainfall data)} \\ \text{(consistency check)} \end{array} \right.$

missing rainfall data :-



$$P_x = \frac{P_1 + P_2 + P_m}{n}$$

(Arithmetic mean method)

$$P_x = \frac{N_x}{n} \left[\frac{P_1}{N_1} + \frac{P_2}{N_2} + \frac{P_m}{N_m} \right]$$

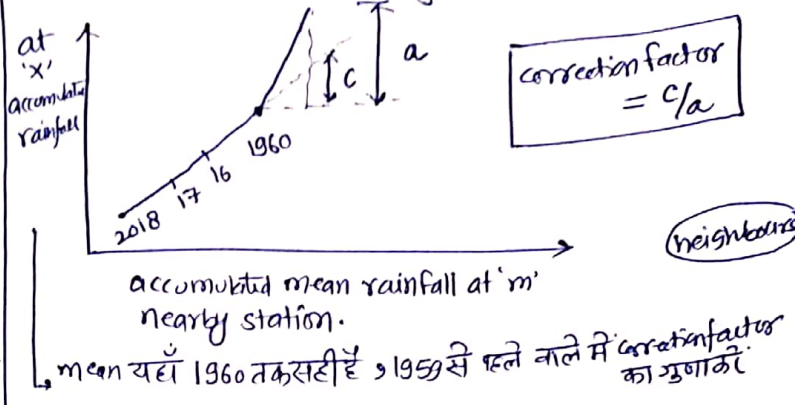
(normal ratio method)

consistency check :- when condition changes $\rightarrow$ data changes of any Rain gauge station.

- (i) shifting of rain gauge
- (ii) construction (like khelgaon) in neighbour
- (iii) change in ecosystem (forest fire)
- (iv) observation error
- (v) replacement of instrument with new one

Double mass curve Technique :- to check inconsistency.

- Plot in reverse chronological order.



Presentation of Rainfall data :-

- (i) mass-curve $\left\{ \begin{array}{l} \Sigma P \\ t \end{array} \right.$ slope = avg. intensity of rain
• By weighing bucket & Natural syphon float type
- (ii) Hyetograph $\left\{ \begin{array}{l} I (\text{cm/hr}) \\ t \end{array} \right.$ • By tipping bucket method
area = rainfall in that period
- (iii) moving Average
aim $\rightarrow$ To analyse trend of rainfall.
Ex. for 5 year moving average $\rightarrow$ avg. rainfall of 1st, 2nd, 3rd, 4th, 5th year-rainfall $\rightarrow$ plot against 3rd year.
- (iv) DAD curve (Depth area duration curve)
aim $\rightarrow$ to convert point rainfall data by R.G. into areal rainfall data.
means It gives areal characteristic of storm.
 $P(\text{of area}) = P_0 e^{-K A^n}$ $K, n \rightarrow$ by regression analysis
max rainfall in catchment at storm centre
- (v) IDF curve (Intensity-duration frequency curve)
aim $\rightarrow$ for risk assessment of dam, bridge, water drainage system.
 $i = \frac{k T^x}{(D+a)^n}$
 $i = \text{Intensity}$
 $T = \text{Return period}$
 $D = \text{duration}$
 $K, a, x, n \rightarrow$ constant for given catchment

maximum Intensity calculation :-

Time since Start (min)	Cumulative rainfall (mm)	rainfall possible in any possible time interval equate			
		30 min	60	90	120 minute
0	0	-	-	-	-
30	6	6-0 = 6	-	-	-
60	18	18-6 = 12	18-0 = 18	-	-
90	21	21-18 = 3	21-6 = 15	-	-
120	28	28-21 = 7	28-18 = 10	-	-

Choose max depth ✓ } together max intensity divide by duration

average rainfall calculation (equivalent uniform depth EUD)

(1) Arithmetic mean method (equival. area method)	- when rainfall variation - nil or uniformly distributed - Least accurate $\bar{P} = \frac{P_1 + P_2 + \dots + P_m}{m}$
(2) Thiessen Polygon 'or' weighted area method	$\bar{P} = \frac{P_1 A_1 + P_2 A_2}{A_1 + A_2}$ $\left\{ \frac{A_1}{A_1 + A_2} = \text{weight factor} \right\}$
(3) Isohyetal method	$\bar{P} = \frac{(P_1 + P_2)}{2} A_1 + \dots$ Isoshyet → line joining points of equal rainfall magnitude. - more rain gauge needed - more accurate method - In this topographic influence is taken into account

Evaporation Reduction :-

(1) Surface area reduction ↓	provide greater depth ∴ SA ↓
(2) mechanical cover	in small lake, small water body construct either permanent roof (canal top roof or solar plant) or Temporary roof.
(3) Application of thin - chemical film cover surface	(1) Hexadecanol (cetyl alcohol) - Best (20-50 % evaporation reduction) - consumption ⇒ 3.5 N/ha/day - evaporation reduction Max. if film pressure maintained @ 4×10^{-2} N/m (2) Stearyl alcohol (octadecanol)

$$\text{Precipitation} = \text{Runoff} + \text{Losses}$$

Losses ⇒ (1) Evaporation + Transpiration
 (2) Infiltration (3) Percolation (4) Interception
 (5) Depression storage

Dalton's evaporation Law :-

$$\text{rate of evaporation (EL mm/day)} \propto e_w - e_a$$

• evaporation takes place when $e_w > e_a$
 • if $e_a > e_w \rightarrow$ condensation will take place.

factors affecting evaporation :-

- (I) Vapour pressure ($e_w \propto e_a$)
- (II) Temp. ($EL \propto \text{Temp}$)
- (III) wind speed $EL \propto \text{wind speed}$
- (IV) atm. pressure $EL \propto \frac{1}{\text{atm. pressure}}$
- (V) water quality $E_d \propto \frac{1}{\text{saline}}$
 • {decrease by 1% by increasing 1% in salinity}
- (VI) surface area $EL \propto SA$
- (VII) Depth of water Body :-
 summer: Shallow depth में $EL \uparrow$ (∵ temp. जल्दी बढ़ जायेगा)
 winter: deep depth में $EL \uparrow$ (temperature कम रहेगी)

measurement of Evaporation by Evaporimeter :-

- (i) Colorado sunken Pan • pan is sunk below ground that water level in pan is at ground level
- (ii) US Geological Survey floating pan • evaporimeter is kept floating in lake (It simulates characteristic of large water body)

	class A Pan	Sunken pan	IS I Pan	US, GS Pan	(CSI)
C_p	0.70	0.78	0.80	0.80	

note:- $\text{Lake evaporation} = C_p \times \text{Pan evaporation}$

estimation of evaporation using empirical formula :-

1- Meyer's formula
(Based on Dalton's Law)

$$E_L = k_m (e_w - e_a) \left[1 + \frac{v_g}{16} \right]$$

$\frac{\text{mm}}{\text{day}}$

$\{ k_m \rightarrow \text{depend on water body} \}$
 $\{ \text{small} \rightarrow 0.50 \text{ large} \rightarrow 0.36 \}$

$e_w = \text{saturation vapour pressure of water surface (mmHg)}$

$e_a \Rightarrow \text{actual vp of overlying air at specified height}$

$v_g \rightarrow \text{monthly mean velocity @ 9m height [km/hr]}$

$$v_h = C h^{1/4} \quad \left(\frac{1}{7} \text{th Law} \right)$$

$\rightarrow \text{valid upto 500 meter}$

note:- ① area with vegetation :- difficult to find -
separately evaporation & transpiration
 $\therefore$ we estimate evapotranspiration.

② area with no vegetation :- hence only evaporation.

Factors affecting evaporation & Transpiration :-

- 1- weather parameter (Temperature, humidity)
- 2- Crop characteristic
- 3- Soil type
- 4- environmental aspect

2- Rohwer's formula

3- Fitzgerald formula

Analytical method to estimate evaporation :-

① Water-Budget method

- simple • Least reliable
- use of hydrological continuity eqn
 $I - O = \Delta S$

② Energy-Balance method

- use of conservation of energy

③ mass-Transfer method

- Based on theory of turbulent mass transfer in boundary layer

Consumptive use C_u :-

evapotranspiration (ET)

water used directly in -
metabolic process of
Building plant tissue

$\downarrow$
 $\therefore$ this term negligible to 'ET' thus neglected.

AET or Evapotranspiration

By Thornthwaite's water balance eqn

lysimeter

field experimental plot or field plot

sensible heat (H_a) :- given by Bowen ratio (β)

ratio of $\left\{ \begin{array}{l} \text{heat} \\ \text{vapour diffusivity} \end{array} \right\}$

Transpiration :- measure by Phytometer

- $\rightarrow$ during day time only.
- $\rightarrow$ water leaves plant body during photosynthesis
- $\rightarrow$ depend on growth period of plant

Isopleth $\rightarrow$ lines on map @ equal depth of ET

PET

Penman eqn

mass transfer energy balance

empirical formula
Blaney criddle formula

$$\frac{\text{cm}}{\text{Season}} \quad \frac{K P}{40} (1.8t + 32)$$

PET :- related to hours of sunshine temp $\rightarrow$ which are taken as measure of solar radiation.

$\left\{ \begin{array}{l} K \rightarrow \text{crop factor} \\ P \rightarrow \text{monthly \% of annual day time hrs (depend on latitude)} \end{array} \right\}$

Infiltration :-

prior to Infiltration , some initial loss :-

1- Interception loss :- (max 2 to 3% of rainfall)

• water capture / interrupted by buildings, leaves, roads, & subsequently goes back to atmosphere by evaporation.

2- Depression storage :-

Empirical Infiltration eqⁿ : ① Horton's eqⁿ

② Soil conservation service practice

③ Antecedent precipitation method

④ Green Ampt model ⑤ Huggins model eqⁿ

Horton's eqⁿ :- $f_t = f_f + (f_i - f_f)e^{-kt}$

$\begin{cases} t=0 & f_t = f_i \\ t=\infty & f_t = f_f \end{cases}$

note:- $\begin{cases} \text{if Intensity of rainfall (i) > } f_c & \text{then } f = f_c \\ i < f_c & \text{then } f = i \end{cases}$

factor's affecting Infiltration (I):-

- I- rain characteristic ② soil characteristic
- ③ surface cover ④ characteristic of infiltrating water

Infiltration Index/indices

index Φ → रेसी बारिश जि-दोनें runoff दिया

① Avg. rainfall above which

rainfall volume = runoff volume

② Avg. rainfall rate during the period of rainfall excess (effective rainfall) (rainfall excess → runoff contribution)

③ initial loss considered as infiltration.

④ if intensity < Φ index then rate of infiltration = rainfall rate

At old max. rate of infiltration = Φ index

⑤ $\Phi \geq W$ index

⑥ if data not given $\Phi = 0.10 \frac{\text{cm}}{\text{hr}}$

W index
① Avg. Infiltration rate during entire period

② initial loss not considered as infiltration.

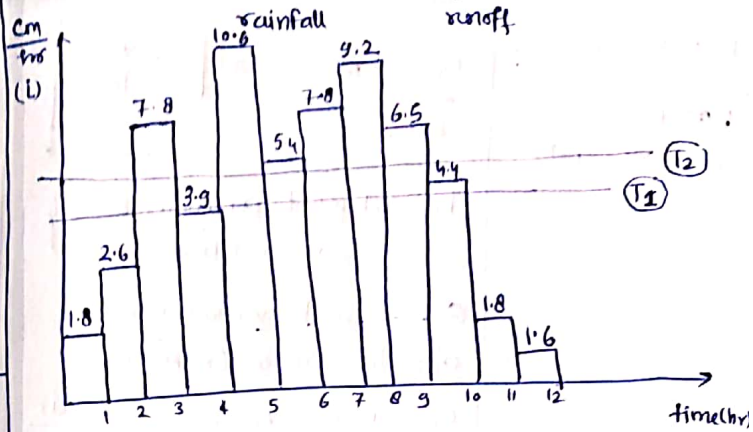
$$W = \frac{P - Q - \Delta l}{t}$$

$\Delta l \rightarrow$ initial loss

Best question :-

$$P = 63.48 \text{ cm} \quad Q = 14.4 \text{ cm}$$

runoff



step-1 :- $W = \frac{P - Q - \Delta l}{t} = \frac{63.48 - 14.4 - 0}{12} = 4.08 \frac{\text{cm}}{\text{hr}}$

step-2 :- Trial-1 $\Phi = 4.08 \frac{\text{cm}}{\text{hr}}$ (assume) = T_1

$$\Phi = \frac{(63.48 - 14.4) - (1.8 + 2.6 + 3.9 + 1.8 + 1.6) \times 1 \text{ hr}}{12 - 5} = 5.32 \frac{\text{cm}}{\text{hr}}$$

(∵ इसमें 4.4 not a period of rainfall excess.)

Trial-2 $\Phi = 5.32 \frac{\text{cm}}{\text{hr}}$ (assume) = T_2

$$\Phi = \frac{(63.48 - 14.4) - (1.8 + 2.6 + 3.9 + 1.8 + 1.6) - 4.4}{12 - 5 - 1} = 5.5 \frac{\text{cm}}{\text{hr}}$$

(now all values considered which gives rainfall.)

Trial-3 :- $\Phi = 5.5 \frac{\text{cm}}{\text{hr}}$

$$\Phi = \frac{32.9 - 5.4}{6 - 1} = 5.5 \frac{\text{cm}}{\text{hr}}$$

hence $5.5 = \Phi$ index $\frac{\text{cm}}{\text{hr}}$

Infiltration measurement :- ① flooding type infiltrometer (infilt)

② rainfall simulator (in lab)

③ by Φ, W index

④ horton curve use

⑤ by hydrograph analysis

Stream flow :- runoff phase of Hydrological cycle.

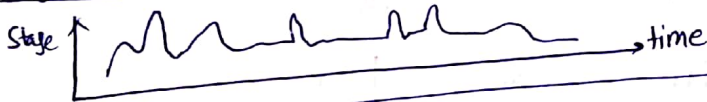
Hydrometry :- science & practice of water measurement.

Stage :- elevation of water above datum

Gauging :- measurement of stage

Stage measurement		
manual Gauging	① Staff Gauge	• Simplest • when single Gauge not sufficient use sectional Gauge. 
	② wire Gauge	• from known elevation - Bridge level 125m 2m 125 - 2 = 123
Automatic stage gauge recorder	① float Gauge recorder	→ mostly used. { to prevent entry of debris & nullify the wave effect → use "stilling well" }
	② Bubble Gauge recorder	record pressure head $P/\rho g = \rho h$

Stage hydrograph :- Plot of stage in chronological order.



Velocity measurement in stream/river :-

① Float method	• It gives (approx velocity) • used for Preliminary survey & small streams in flood. $V_{avg} = \frac{K V_s}{0.85-0.95}$ → distance by float / time
② Current meter	$V = a N_s + b$ (vertical axis → (Price current) meter) → rotating disc provided on vertical axis • to use it major component of velocity in longitudinal direction.
③ Calibration of Current meter → By Towing Tank	- Single point formula $V_{avg} = V_0 \cdot 87$ - double (2 point) formula $V_{avg} = \frac{V_0 \cdot 27 + V_0 \cdot 87}{2}$ 3 point velocity formula $\Rightarrow \frac{V_0 \cdot 27 + V_0 \cdot 87}{2} + V_0 \cdot 67$
④ Wading → Process of finding velocity in - shallow streams	

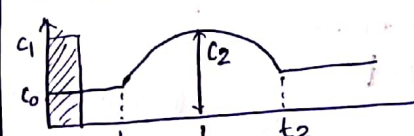
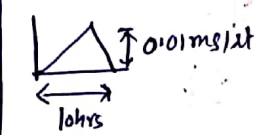
Streamflow or discharge measurement

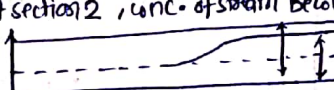
direct method

- 1- Area velocity method
- 2- moving boat "
- 3- Dilution Technique
- 4- electromagnetic method (for low)
- 5- ultrasonic method

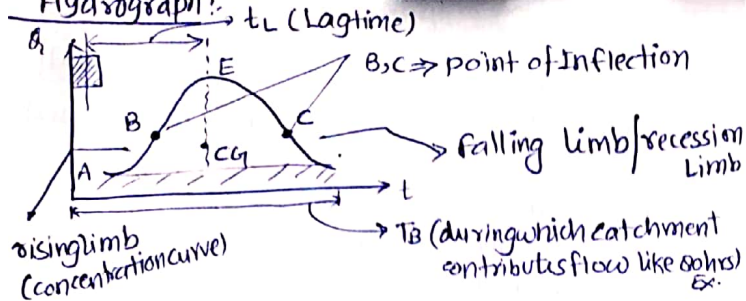
indirect method

- 1- slope area method
- 2- By hydraulic str. (weir, flumes, gated structures)

① Area velocity method	• solid section method • standard current-meter method • Depth at various verticals using sounding rod
② moving boat method	• For wide stream • depth by <u>ecodepth recorder</u> • Current meter installed in Boat
③ Dilution Technique (Based on mass-conservation)	• use soln of stable chemical (NaCl, Sodium dichromate, coloride)
④ sudden Injection / Gulp / Integration method	• tracer introduced all at once.  $C_0 \Rightarrow$ background conc. (tracer initial concentration) $Q = \frac{V_1 C_1}{\int_{t_1}^{t_2} (C_2 - C_0) dt} = \frac{\text{wt of tracer introduced in Gulp}}{\text{area of time concentration curve}}$ Ex. $V_1 = 60 \text{ litre}$ $C_1 = 250 \text{ mg/lit}$ $V_1 C_1 = 147.5$  $\text{Mea} = \frac{1}{2} \times (10 \times 3600) \text{ sec} \times 0.01 \times 10^{-6} \text{ mols} \times 9.81 \text{ m/s}^2$

⑤ constant rate Injection method (plateau method)	• tracer introduced at constant rate. $C_1 \rightarrow$ added continuously @ Q_1 rate at section 1 - Let initial concentration of tracer in stream = C_0 - at section 2, conc. of stream becomes constant = C_2  $Q_1 C_1 + Q_0 C_0 = (Q_1 + Q_0) C_2$ $Q = \frac{Q_1 (C_2 - C_0)}{(C_2 - C_0)}$ 120 ppm = 120 unit / 106
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Hydrograph:



- 1- lagtime (t_L) $\Rightarrow$ between CG of rainfall to CG of hydrograph
- 2- Rising limb AB $\Rightarrow$ slope depend on rainfall & catchment characteristics.
- 3- falling limb $\Rightarrow$ ————— only catchment "
- 4- crest segment BEC $\Rightarrow$ ends at point of inflection [condition of max. storage]
- 5- Q_t (recession limb) $\propto$ storage remaining at that time

Shape of Hydrograph depend:

climatic factor	Physiographic factor
1- rainfall duration, time, intensity	(i) Basin characteristic
2- Evapotranspiration	(ii) Infiltration "
3- Initial Loss	(iii) channel "

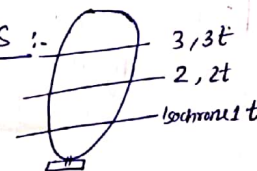
factors affecting Hydrograph:

1- Shape of catchment	
Size of catchment	$Q \propto A^n$
(3) Slope of catchment	$\theta \uparrow \Rightarrow$ dispose runoff faster
(4) drainage density	$\uparrow \Rightarrow$ peak $\uparrow$
(5) land use	Vegetation & forest reduce peak due to retardation of overland flow
(6) urbanisation	$\therefore$ infiltration $\downarrow$ $\therefore$ runoff $\uparrow \Rightarrow$ peak $\uparrow$ $\therefore$ time req. less
(7) effect of rainfall	intensity, duration, effect of aerial distribution over catchment, direction of storm movement
(8) Soil moisture condition	

Some Definitions:

(i) Form Factor	$(FF) = \frac{B_{avg}}{L_{axial}}$ $FF \uparrow$ early Peak
(ii) Compactness coefficient (C_c)	$C_c = \frac{P_{basin}}{P_c}$ $A = \pi r^2 = \text{catchment area}$ $P_c = 2\pi r$ $C_c = 1$ (Circular) shape catchment $C_c > 1$ PT $\Rightarrow$ (fern) shape " TB $\uparrow$, late peak, low peak discharging $C_c < 1$ PT $\Rightarrow$ (fan shape) TB $\downarrow$ early peak
(iii) Stream density	total no. of streams per unit area of catchment $SD \uparrow \Rightarrow$ early peak, less Base period
(iv) Drainage Density	length of streams per unit area $DD \uparrow \Rightarrow$ early peak, less Base period

Isochrones:



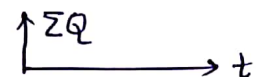
- Imaginary lines across catchment from where particles of water takes same time to reach outlet.

Flow duration curve / discharge frequency curve:

Plot of discharge Q Vs % of time flow was equalled or exceeded

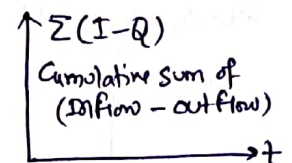


(ii) Flow mass curve:

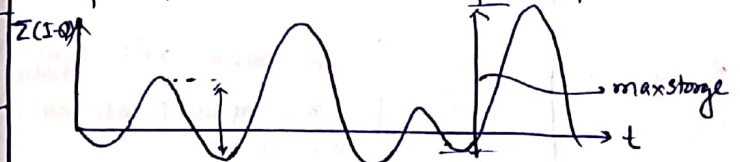


(iii) Sequent peak logarithm:

To find storage capacity of reservoir



• Sequent peak $\rightarrow$ Greater than previous peak.

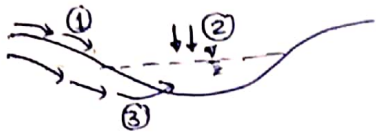


max Difference b/w This sequent peak & lowest trough during period under study $\rightarrow$ Storage capacity of reservoir.

• Runoff $\Rightarrow$ flow away of precipitation from catchment

• Runoff types :-

1. direct runoff or direct storm-runoff or storm runoff
- enters in stream just after precipitation
 - (i) flow over land surface (surface runoff)
 - (ii) direct precipitation (directly on stream surface)
 - (iii) Prompt interflow



- (ii) Baseflow or Groundwater flow or dry weather flow or effluent seepage
- delayed flow (result of previous rain) which enters the stream essentially as ground water flow

$$[\text{Total Runoff} = \text{direct runoff (DR)} + \text{Baseflow (B)}]$$

surface runoff

overland flow

(flow of water over land before joining open channel)

- Length $\rightarrow$ small
- flow $\rightarrow$ Laminar

open channel flow

flow turbulent

Runoff estimation :-

1- using runoff coefficient ($R = Kp$)

2- rational method

$$Q_p = \frac{1}{36} \cdot u_i \cdot A \quad \begin{matrix} \text{cm/s} \\ \downarrow \\ \text{m}^3/\text{s} \end{matrix} \quad \begin{matrix} \text{km}^2 \\ \downarrow \\ \text{ha} \end{matrix}$$

3- infiltration curve

4- Empirical formula

Dicken's formula

$$Q_p = C_D \cdot A^{3/4} \quad \begin{matrix} \text{m}^3/\text{sec} \\ \downarrow \\ \text{km}^2 \end{matrix} \quad \begin{matrix} (6-30) \end{matrix}$$

Cyve's formula

$$Q_p = C_r A^{2/3}$$

Inglis formula

$$Q_p = \frac{124A}{\sqrt{A+10.4}} \approx 123\sqrt{A}$$

5. by using water budget eqⁿ

6- By hydrograph

7- By ϕ index

8- Rainfall-runoff correlation

Runoff- Rainfall relationship :-

$$R = aP + b$$

$$b = \frac{\sum R - a \sum P}{n}$$

of $R < 1$

$$a = \frac{n \sum PR - \sum P \sum R}{n \sum P^2 - (\sum P)^2}$$

$$r = \frac{n \sum PR - \sum P \sum R}{\sqrt{[n \sum P^2 - (\sum P)^2][n \sum R^2 - (\sum R)^2]}}$$

$$\text{(i) Rainfall excess (n)} = \underbrace{\text{total rainfall } P}_{\text{(n)}} - \underbrace{\text{initial loss } \Delta l}_{\phi \cdot t} - \underbrace{\text{Infiltration } I}$$

(ii) Surface runoff will exist where there is rainfall excess

(iii) Effective rainfall :- portion of rainfall which cause direct runoff. (surface runoff + prompt interflow) (slight more than rainfall excess)

• effective rainfall we got by **DRH**

$$\text{effective rainfall} = \frac{\text{DRH volume}}{\text{catchment area}}$$

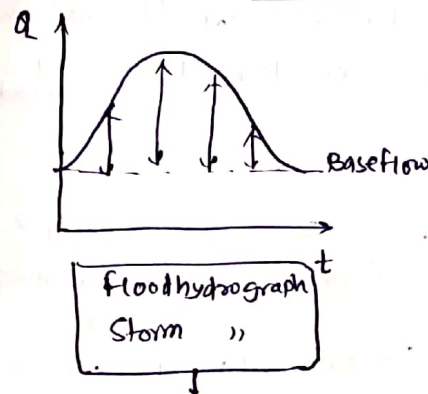
• interflow $\rightarrow$ small $\therefore$ direct runoff = surface runoff
hence rainfall excess & effective rainfall used synonymously.

(iv) effective rainfall Hyetograph (ERH) :- rainfall Hyetograph - infiltration loss - initial loss

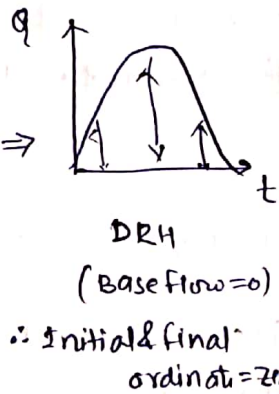
note: DRH is result of ERH (eff. rainfall)

$$\text{(v) direct runoff depth (n)} = \frac{\text{direct runoff volume}}{\text{catchment area}}$$

$$n(\text{cm}) = \frac{0.36 \times \sum O_i (\frac{\text{m}^3}{\text{sec}}) \times t_i (\text{hr})}{A (\text{km}^2)}$$



It includes Baseflow



DRH
(Baseflow = 0)
 $\therefore$ Initial & final ordinates = zero

Unit Hydrograph :- by Sherman (1932)

- ① DRH which results in runoff depth (n) = 1 cm
- ② DRH $\xrightarrow{n=1\text{cm}}$ UH
- ③ ~~DRH~~ DRH UH $\rightarrow$ intensity = $\frac{1}{D} \frac{\text{cm}}{\text{hr}}$

UH assumption :-

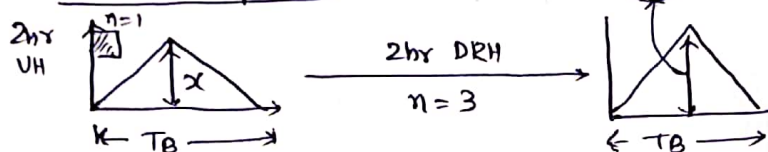
(i) Time Invariance :-

for same catchment $\begin{cases} 2015 & \text{DRH } T=2\text{hr } n=1 \\ 2020 & \text{DRH } T=2\text{hr } n=1 \end{cases}$

मतलब

shape of DRH for given eff. rainfall = constant.

(ii) Linear Response :-



(iii) rainfall ~~at~~ $\rightarrow$ uniform distribution

(iv) rainfall intensity $\rightarrow$ constant

(v) runoff obtained due to rainfall only (not snow fall)

- note:-
- ① $Q_{\text{peak}}(2\text{hr}) > Q_{\text{peak}}(4\text{hr})$
 - ② TB of 2hr UH $\Rightarrow 48\text{hr}$
Then $\rightarrow 4\text{hr UH} \Rightarrow 48 + 2 = \text{TB} + 2\text{hr}$
 - ③ Area of 2hr UH = 4hr UH = Q_{UH}

($n=1\text{cm}$ constant)

'or' sum of ordinates are same for all UH

Use of unit hydrograph :- ① develop flood hydrograph which is used in design of hydraulic str.

- ② development of Flood forecasting, flood warning system
- ③ Extension of flood flow Records based on rainfall records

Unit hydrograph limitation :- ① $2 \leq A \leq 5000 \text{ km}^2$

② only for rain precipitation

(not snow)

③ Catchment should not have unusual pond, Lake.

④ Precipitation should not be nonuniform.

Shydrograph :- DRH from continuous effective-rainfall of intensity = $\frac{1}{T_0} \text{ cm/hr}$

Q_{eqb} or Q_{max}

obtained if full rainfall converted into discharge

$\lim_{t \rightarrow \infty} Q(t) = Q_{\text{eqb}}$

$Q_{\text{eqb}} = \text{area} \times \text{intensity} \left(\frac{1 \text{ cm}}{T_0 \text{ hr}} \right)$

$Q_{\text{eqb}} = 2.78 \frac{A}{T_0} \rightarrow \text{km}^2$
 $\rightarrow \text{m}^3/\text{sec.}$

$\frac{dQ}{dt} = 0 \quad t = ?$

Put t in $Q(t) \Rightarrow Q_{\text{eqb}}$

Best question :- note:- यदि 6hr UH दिया है तो 6hr के पहले के सारे ordinates & curve के भी same वही ordinate होंगे।

Ex. How to make 2hr UH $\rightarrow$ 3hr UH

यहाँ पर $m = 3/2 = 1.5$

T	2hr UH ordinate	S-curve addition	S-curve ordinates	offset curve (Lag by 3hr) जितने का व. ग. हो उतने से lag कराओ	Δy	3hr UH $\uparrow$ ordinate $\left(\frac{\Delta y}{m} \right) \rightarrow 1.5$
0	0		0		0	0
1	3		3		3	2
2	8	0	8		8	5.33
3	6	3	9	0	9	6
4	3	8	11	3	8	5.33
5	2	9	11	8	3	2
6	0	11	11	9	2	1.33

यहाँ पर Q_{eqb} or Q_{max} of S-curve $\Rightarrow 11 \text{ m}^3/\text{sec}$ which occurred at $T=4\text{hr}$.

Synthetic unit hydrograph :- $\Rightarrow$ for ungauged basin

• select 3 parameters

① Basin lag time (t_p) :- mid point of rainfall excess to peak of UH

$$(t_p) = C_t (L \cdot L_{ca})^{0.30} \text{ hrs.}$$

② Base time T_B :- $T_B = 72 + 3t_p$

Std. duration of rainfall as per Snyder

$$t_r = \frac{t_p}{5.5} \rightarrow \text{check whether equal to given for UH or not}$$

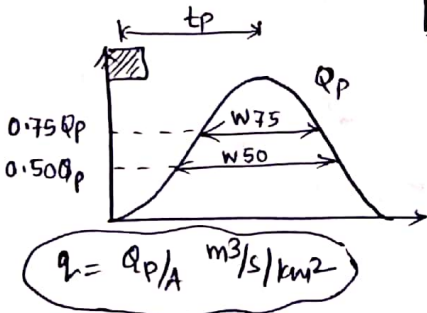
if not $\rightarrow$

then $t_p' = \frac{21}{22} t_p + \frac{T_0}{4}$ मिने हर का बराबर

'or' $t_p' = t_p + \left(\frac{T_0 - t_r}{4} \right)$

③ Peak discharge Q_p :-

$$Q_p = 2.78 C_p \frac{A}{t_p'}$$



$$W_{50} = \frac{5.07}{9.108}$$

$$W_{75} = \frac{W_{50}}{1.75}$$

Instantaneous UH :- $\lim_{T \rightarrow 0} \text{Thr UH} \rightarrow \text{IUH}$ (rainfall intensity $\rightarrow \text{max}$)

$$\text{runoff depth of IUH} = \frac{\int_0^\infty u(t) dt}{A} = 1 \text{ cm always}$$

IUH advantage :- ① Independent of duration of rain

② Independent of rain characteristics (depend on catchment characteristics)

③ Slope of 'S' curve { intensity = 1 cm/hr } = ordinate of IUH
 $u(t) = \frac{1}{t} \frac{ds}{dt}$ if $i = 1 \text{ cm/hr}$

① Clark model :- uses concept of time-area histogram

$\rightarrow$ IUH obtained by routing rainfall excess.

② Nash model (IUH) :- using routing through a cascade of linear reservoir in linear reservoir Storage & Q (outflow)

Exceedence frequency :- इससे ज्यादा या बराबर कितनी value है।

Q arrange rank wise (m) दो same तो ज्यादा वाली rank मिले

$T = N/m$ $P = 1/T$ • draw peak discharge Vs Prob. curve
California method

1- Weibull method (plotting position method)
 $\rightarrow$ It gives best result

$$P = \frac{m}{N+1}$$

2- California method

$$P = \frac{m}{N}$$

3- Hazen

$$P = \frac{m-0.5}{N}$$

4- US corps (absurd result)

$$P = \frac{N+1}{m} \text{ (Weibull का 3वां)} \quad \left(\begin{array}{l} P \rightarrow \text{occurrence prob.} = \frac{1}{T} \\ P+q=1 \end{array} \right)$$

$$P = n_c r^q q^{n-r} \quad \left\{ \begin{array}{l} P \rightarrow \text{occurrence prob.} = \frac{1}{T} \\ P+q=1 \end{array} \right.$$

reliability = q^n
 risk = $1 - q^n$
 $\rightarrow$ Particular event never happens ($r=0$) in given time span of n years.

Particular event happens at least once or exceeds once (not exactly $\rightarrow$ once)

for longer Return period (like $T=500$ years)
 Theoretical prob. distribution used $\rightarrow$

- 1- Gumbel method $\rightarrow$ flood peak, max. rainfall, max. wind speed
- 2- log pearson type-3 distribution
- 3- log normal distribution

Grumble method :-

$$P(x \geq x_0) = 1 - e^{-e^{-y}}$$

$y \rightarrow$ dimensionless variable

$$y = \frac{1.285(x - \bar{x})}{s_x} + 0.577$$

$$x_T = \bar{x} + k_T s_{n-1}$$

Peak discharge $\downarrow$ mean $\downarrow$ S.D $\downarrow$ frequency factor

$$k_T = \frac{y_T - \bar{y}_n}{s_n} \rightarrow 0.577$$

1.285

$$y_T = -\ln \ln \left[\frac{T}{T-1} \right]$$

reduced variate

for $N \geq 50$ no. of year record
 $\bar{y}_n = 0.577$
 $s_n = 1.285$

Asper Grumble :- value of flood with return period $T = 2.33$ year is called mean annual flood.

• due to limited data, Grumble method has some error in x_T $\therefore$ use confidence limit

$$\begin{cases} x_1 = x_T + f(c) s_e \\ x_2 = x_T - f(c) s_e \end{cases}$$

$f(c) \rightarrow$ fn of confidence limit

$s_e \rightarrow$ Prob error

C. %	f(c)
68	1
95	2

$$s_e = \frac{s_{n-1}}{\sqrt{N}} \rightarrow \sqrt{\frac{\sum (x - \bar{x})^2}{N-1}}$$

$$b = \sqrt{1 + 1.3 k_T + 1.1 k_T^2}$$

Type of flood :-

① Standard project flood (SPF)

$\approx 40-60\%$ of MPF

note:- used where failure of str. will cause less severe damage.

• flood from most severe combination of meteorological & hydrological condition.

• [But excluding extremely rare combination]

② max. prob. flood (MPF)

note:- failure of str. will cause loss of life and catastrophic damage

• SPF + extreme rare combination or condition + catastrophic flood

③ probable max. Precipitation (PMP)

$$PMP(m) = 42.16 D^{0.475}$$

$\rightarrow$ hr

→ Greatest of extreme rainfall of given duration physically possible over an area.

④ Design flood

- adopted in design
- can be SPF, MPF or any other value based on desired return period, degree of flood protection offered, cost of construction of str.
- choose cost/benefit $\Rightarrow$ min

note:- Culvert (design flood) :-

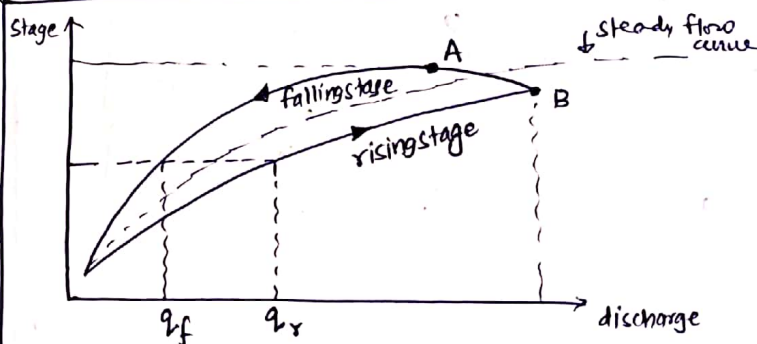
- obtain from statistical consideration, say a flood of 50 year return period.

Time of concentration $\rightarrow$ by "Kirpich eqn"

$$t_c (\text{min.}) = 0.01947 L^{0.77} S^{-0.385}$$

max. length of travel of water (m) $\rightarrow \frac{\Delta H}{L}$

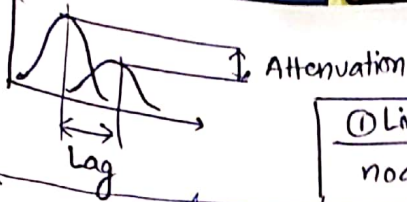
$\Delta H \Rightarrow$ difference in elevation b/w most remote point on catchment & outlet.



A $\rightarrow$ max. stage point

B $\rightarrow$ max. discharge point

(Stage-discharge hysteresis)



① Linear channel :-
no attenuation, only lag

Linear Reservoir :-
Storage $\propto f^n$ (outflow discharge Q)
Such storage $\rightarrow$ linear storage
 $\rightarrow$ put $x=0$ in Muskingum eqn
 $S = f^n(Q)$

outflow from reservoir

Uncontrolled

Controlled

• Peak of outflow hydrograph will occur at point of intersection of inflow and outflow hydrograph
ie inflow discharge = outflow discharge
Ex. freely operating spillway.

will not max. at intersection.

equation used in flood routing

continuity equation

$$I - Q = \Delta S$$

$$\left(\frac{I_1 + I_2}{2}\right) \Delta t - \left(\frac{Q_1 + Q_2}{2}\right) \Delta t = S_2 - S_1$$

St. Venant's eqn for unsteady flow

continuity eqn of unsteady flow

momentum eqn

$$\frac{\partial Q}{\partial x} + T \frac{\partial y}{\partial t} = 0$$

$$\frac{\partial y}{\partial x} + \frac{y}{g} \frac{\partial v}{\partial x} + \frac{1}{g} \frac{\partial v}{\partial t} = S_0 - S_f$$

Flood routing

Hydraulic routing

(distributed routing)

$$Q = f^n(x, t)$$

space time

• complex method

continuity eqn

eqn of motion of unsteady flow (St. Venant's eqn) or momentum eqn.

Hydrologic routing

(Lumped routing)

$$Q = f^n(\text{time})$$

• simple method

only continuity eqn used
 $I - Q = \Delta S$

Hydrologic routing

Hydrologic channel routing

$$S = f^n(I, Q)$$

Ex. Muskingum method

Hydrologic reservoir routing

or Storage routing

or Level pool routing

$$S = f^n(Q)$$

Graphical method

Numerical method

- 1- modified Puls method
- 2- Gracich method
- 3- std. 4th order Runge-Kutta method

Hydrologic channel routing :- $S = f^n(I, Q)$

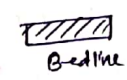
$$\text{Total storage (S)} = \text{Prism storage (S}_p\text{)} + \text{Wedge storage (S}_w\text{)}$$

$$\Downarrow \qquad \qquad \qquad \Downarrow$$

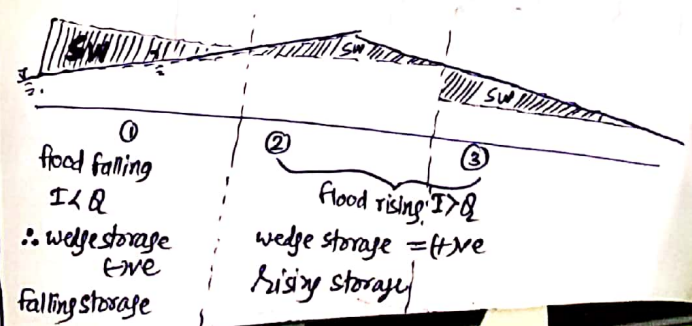
$$f^n(Q) \qquad \qquad \qquad f^n(I)$$

Prism storage :- when Bed line parallel to water surface

(Uniform) flow



Wedge storage :-



Muskingham method :-

(Hylogic channel routing)

$$S = f^n(I, Q)$$

$$S = k \left[x I^m + (1-x) Q^m \right]$$

k = storage time constant (dimension of time)

x = weightage factor $0 \leq x \leq 0.5$

$$S = f^n(Q)$$

Storage depend
on outflow
Ex. linear reservoir

$$S = f^n(I, Q)$$

Storage equally
depend on
 I & Q .

for given reach
 $x, k \rightarrow$ constant

m = constant exponent

$$m = 0.6$$

artificial rectangular channel

$$m = 1$$

for natural channel

$$\left(\frac{I_1 + I_2}{2} \right) \Delta t - \left(\frac{Q_1 + Q_2}{2} \right) \Delta t = S_2 - S_1 = \Delta S$$

$$Q_n = C_0 I_n + C_1 I_{n-1} + C_2 Q_{n-1} \quad \left\{ \begin{array}{l} n_{\min} = 2 \end{array} \right.$$

$$Q_2 = C_0 I_2 + C_1 I_1 + C_2 Q_1 \quad \left\{ \begin{array}{l} \text{assume } Q_1 = I_1 \end{array} \right.$$

$$C_0 + C_1 + C_2 = 1$$

$$\Delta t \in (2kx, k)$$

$\Delta t \downarrow$ small $\Rightarrow$ more
interval more calculation

$$\left\{ \begin{array}{l} C_0 = \frac{0.5 \Delta t - kx}{k(1-x) + 0.5 \Delta t} \\ C_1 = \frac{0.5 \Delta t + kx}{k(1-x) + 0.5 \Delta t} \\ C_2 = 1 - C_0 - C_1 \end{array} \right.$$

To The Point
Hyd
21/4/2020